INFO 510 Bayesian Modelling and Inference

Lecture 15 – Hamiltonian (Hybrid) Monte Carlo

(MCMC algorithm)

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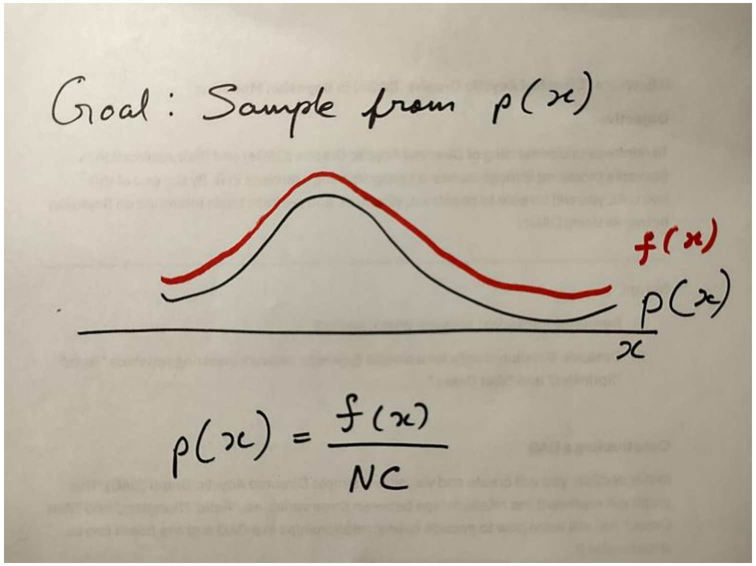
Metropolis-Hastings



We don't always know the exact form of p(x)

We only know this numerator term f(x)

Can we use only f(x) to get samples from p(x)?

Idea behind MCMC

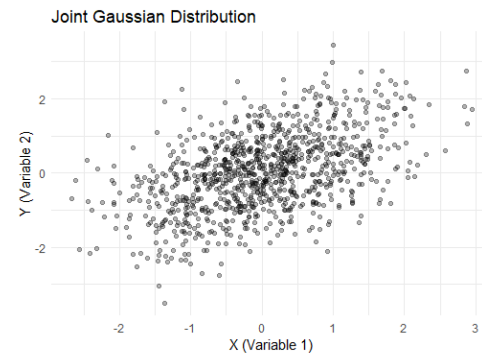
$$\underbrace{X_0 \rightarrow X_1 \rightarrow X_2}_{\text{Burn-in}} \rightarrow \rightarrow \rightarrow \underbrace{X_B \rightarrow X_{B+1}}_{\text{Samples from } p(x)}$$

The accept-reject step

calculate the acceptance probability, $A(x_n \rightarrow x^*)$,

$$A(x_n \to x_*) = \min\left(1, \frac{p(x^*)}{p(x^n)} \times \frac{q(x_n|x^*)}{q(x^*|x_n)}\right)$$

Gibbs Sampling

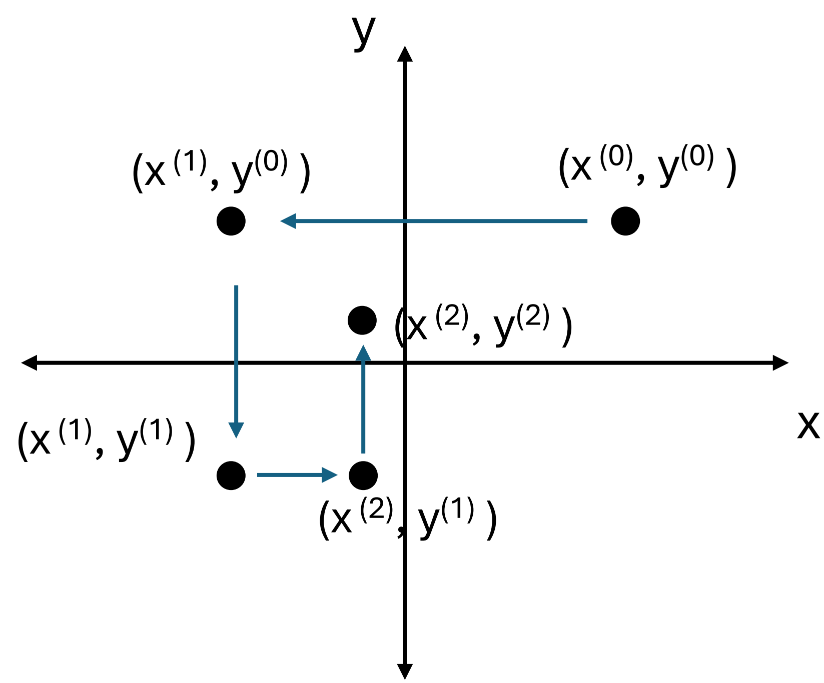


Start at $x^{(0)}, y^{(0)}$

Sample $x^{(1)} \sim P(x^{(1)} \mid y^{(0)})$

Sample $y^{(1)} \sim P(y^{(1)} \mid x^{(1)})$

Sample $x^{(2)} \sim P(x^{(2)} \mid y^{(1)})$



Keep alternating between x and y, each time sampling from their conditional distributions.
Eventually, the sequence of samples will converge to the true joint distribution.

Metropolis Hastings (limitations)

MH often exhibits random walk behavior, particularly in high dimensions
- Slow exploration of parameter space, high autocorrelation

Every proposed move requires computing the acceptance probability

Rejected proposals waste computation while the chain stays in place

High rejection rates mean longer burn-in periods

Gibbs sampling (limitations)

Problems sampling if PDFs are completely separate

Can be slow if dimensions are highly correlated

Hamiltonian Monte Carlo (Hybrid Monte Carlo)

HMC improves upon standard MCMC methods by using information from the gradients of the probability distribution to propose better next steps in the sampling process, allowing more efficient exploration of the parameter space

As the number of dimensions increases, traditional MCMC methods struggle to explore the posterior space efficiently

HMC leverages the structure of the posterior distribution by using concepts from physics

It borrows from Hamiltonian dynamics to explore the posterior efficiently, avoiding the slow random-walk behavior of Metropolis-Hastings

It is an advanced MCMC method used in statistical sampling, and it's particularly efficient for exploring high-dimensional spaces

Posterior Density: $P(\theta|y)$

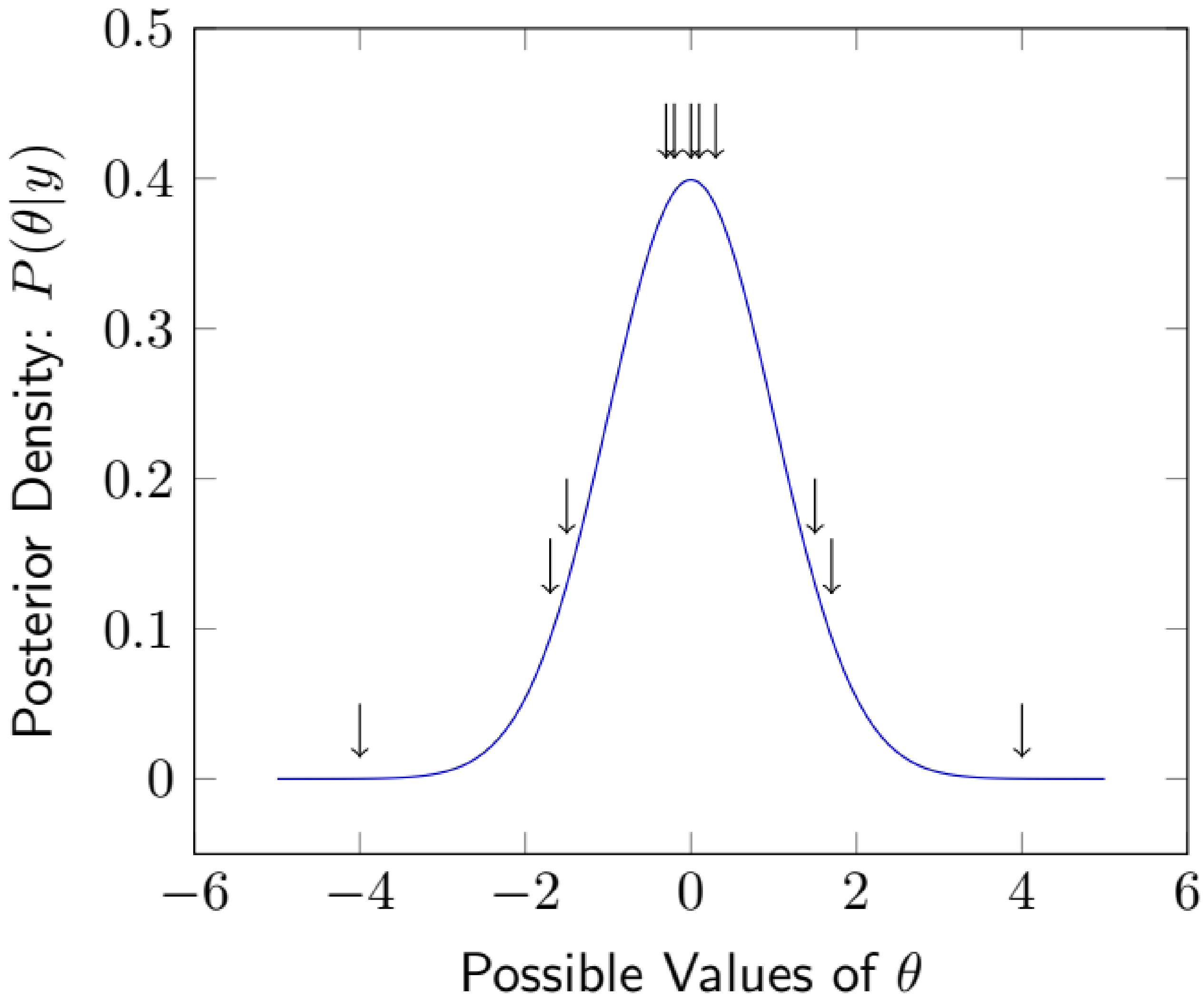
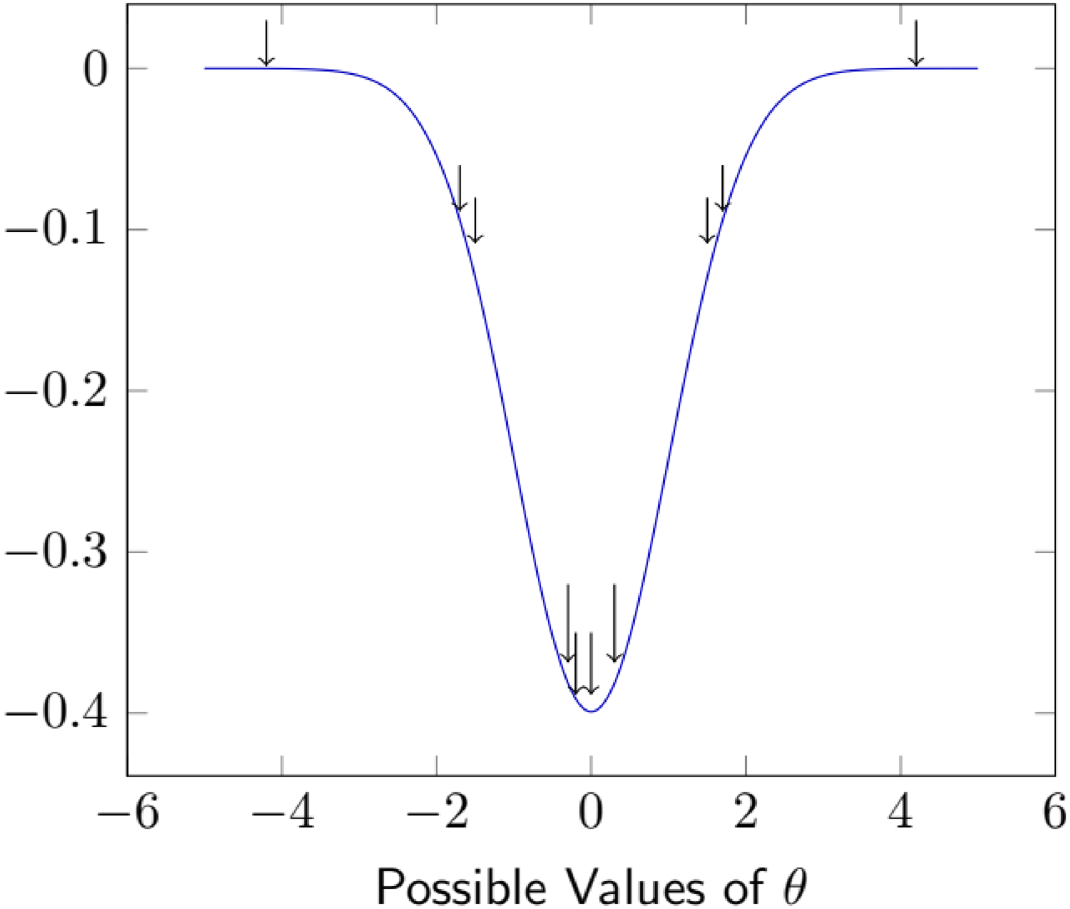


Image courtesy: Carolyn Anderson, University of Illinois

-log(Posterior Density): $-\log(P(\theta|y))$



$$\nabla \log p(\theta \mid y) = \nabla \log p(y \mid \theta) + \nabla \log p(\theta)$$

The gradient of the log-posterior

Image courtesy: Carolyn Anderson, University of Illinois

Statistical Mechanics: movement over fixed time period

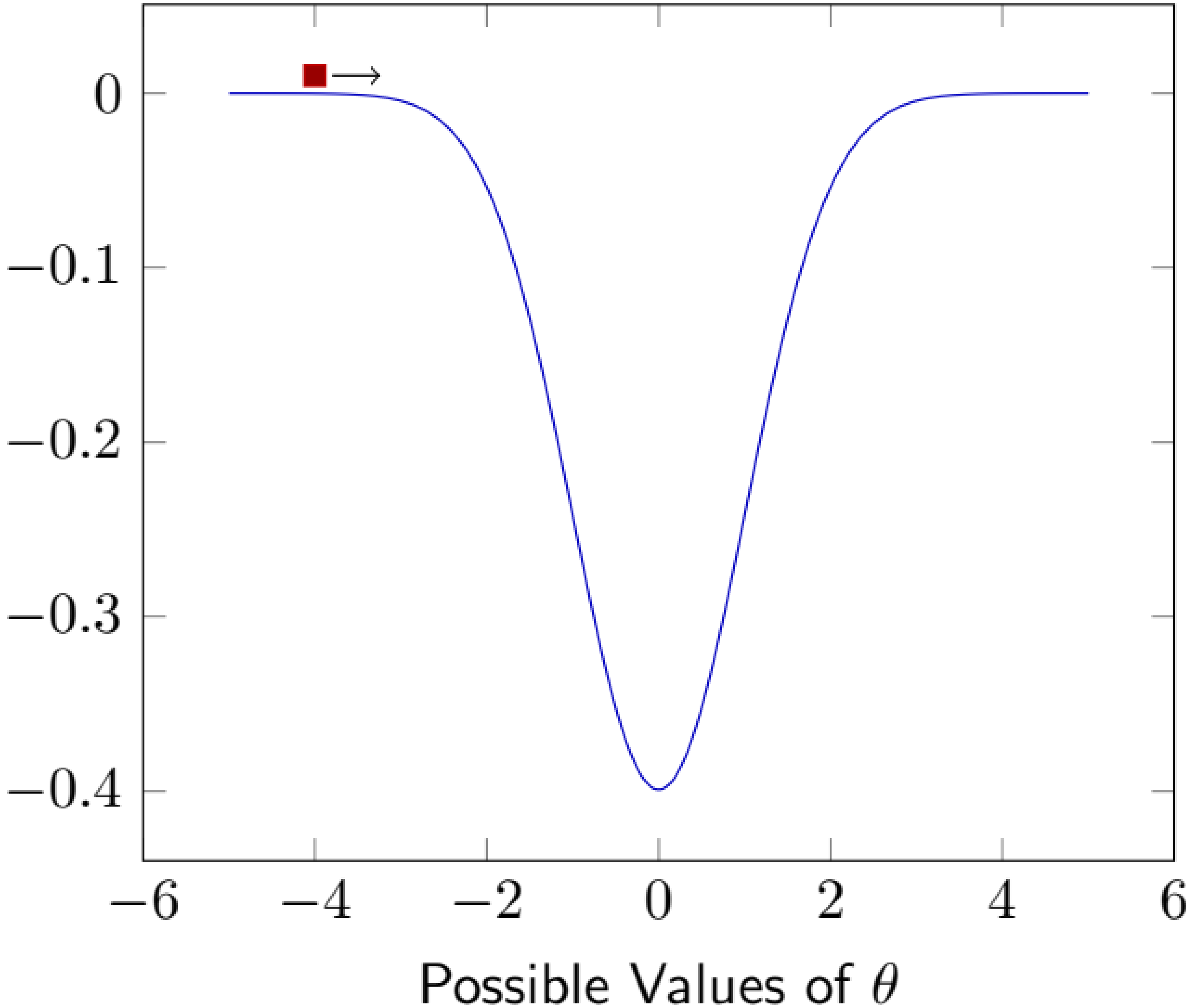


Image courtesy: Carolyn Anderson, University of Illinois

Hamiltonian Dynamics

In physics, the Hamiltonian of a system is the total energy, which is the sum of kinetic energy and potential energy

In Hamiltonian mechanics, the state of a system is described by two things –

Position q - Where the particle is (e.g. in the probability space)

Momentum p - How fast the particle is moving (this controls how much energy it has to "move" in the space)

Hamiltonian function - H(q,p) describes the total energy of the system

For a typical particle system, the Hamiltonian looks like

$$\mathbf{H(q,p) = U(q) + K(p)}$$

Hamiltonian Dynamics

For a typical particle system, the Hamiltonian looks like

$$\mathbf{H(q,p) = U(q) + K(p)}$$

q is the position of a particle

p is the momentum

U(q) – potential energy based on the position, which can be thought of as the energy landscape of the probability distribution we are trying to sample from

K(p) – Kinetic energy (energy due to motion or momentum)

Mathematical Formulation of HMC

HMC introduces an auxiliary variable p (momentum) for each parameter q (position)

The goal is to simulate the motion of a particle with position **q** and momentum **p** under the influence of a potential energy function given by the negative log-posterior

$$U(q) = -\log P(\theta|D)$$

The Hamiltonian equations of motion describe how q and p evolve over time

$$\frac{dq}{dt} = \frac{\partial H}{\partial p}, \qquad \frac{dp}{dt} = -\frac{\partial H}{\partial q}$$

velocity of the particle (or how fast the generalized coordinate is changing over time) is given by the rate of change of the Hamiltonian with respect to the momentum

the rate of change of momentum over time is driven by the gradient of the Hamiltonian with respect to the generalized coordinate q

The Hamiltonian equations of motion describe how q and p evolve over time

$$\frac{dq}{dt} = \frac{\partial H}{\partial p}, \qquad \frac{dp}{dt} = -\frac{\partial H}{\partial q}$$

H = Hamiltonian

Standard nomenclature
q = θ; p = m

velocity of the particle (or how fast the generalized coordinate is changing over time) is given by the rate of change of the Hamiltonian with respect to the momentum

the rate of change of momentum over time is driven by the gradient of the Hamiltonian with respect to the generalized coordinate q

By using these equations, we can typically explore our joint parameter space (θ) much more quickly than via any other MCMC method

For Bayesian analysis it can be viewed as joint probability density function

Hamiltonian $H(\theta, m) = U(\theta) + K(m)$

U is potential energy (Posterior density)
K is Kinetic energy (momentum variables)

Alternately, it can be viewed as a joint pdf $\mathbf{P(\theta, m) = P(\theta|data) \times P(m)}$

$$\log(P(\theta, m)) = \underbrace{\log(P(\theta|data))}_{\text{Posterior}} + \log(P(m)) \qquad = -H(\theta, m)$$

$$\underbrace{\log(P(\theta|data))}_{\text{Posterior}} \propto \underbrace{\log(P_{prior}(\theta) \,.\, P(data|\theta))}_{\text{Prior x likelihood}} \qquad = -U(\theta)$$

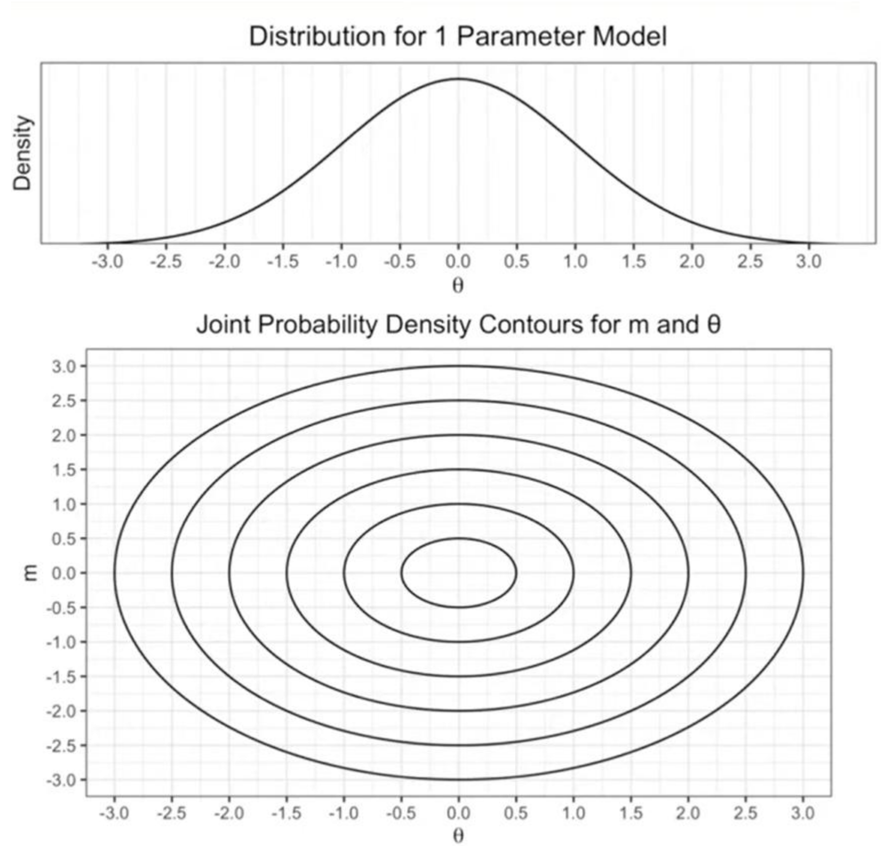
$$\log(P(m)) \propto \log(P(m)),\ P(m) \sim MVN(0, \Sigma) \qquad = -V(m)$$

One parameter model with one momentum variable

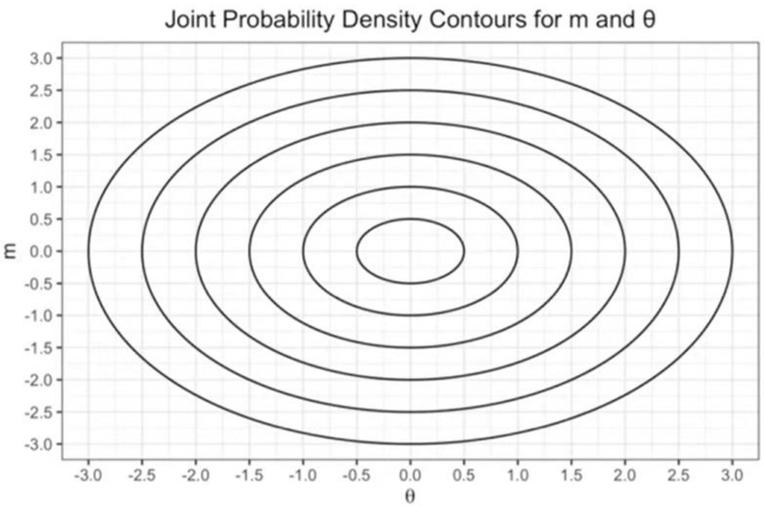
Our single parameter **θ ~ N(0, 1)** (wish to obtain MCMC samples using HMC)

A single momentum variable,
m ~ N(0, 1)

We can show the contours of the joint PDF for **θ and m;** sometimes called “**phase space**”

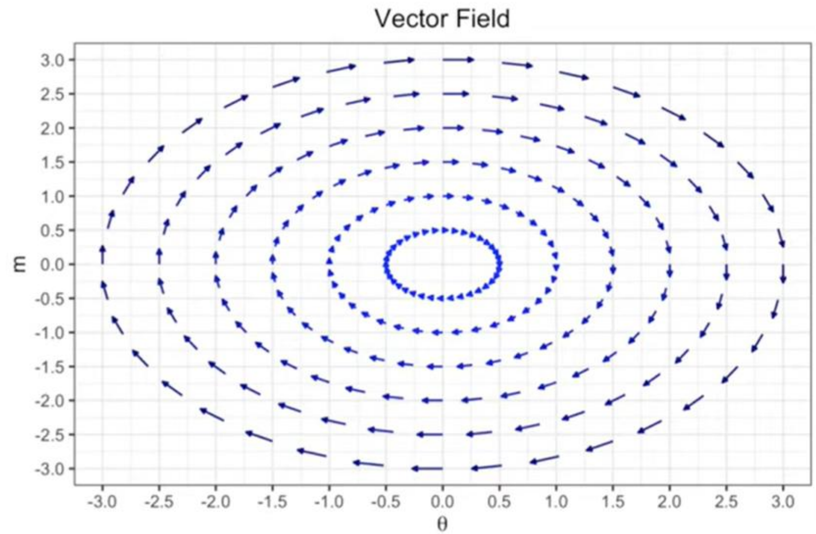


Hamilton’s equation allows us to travel around this contours



$$\frac{d\theta}{dt} = \frac{\partial H}{\partial m}$$

$$\frac{dm}{dt} = -\frac{\partial H}{\partial \theta}$$



The two diff. equations creates the vector fields

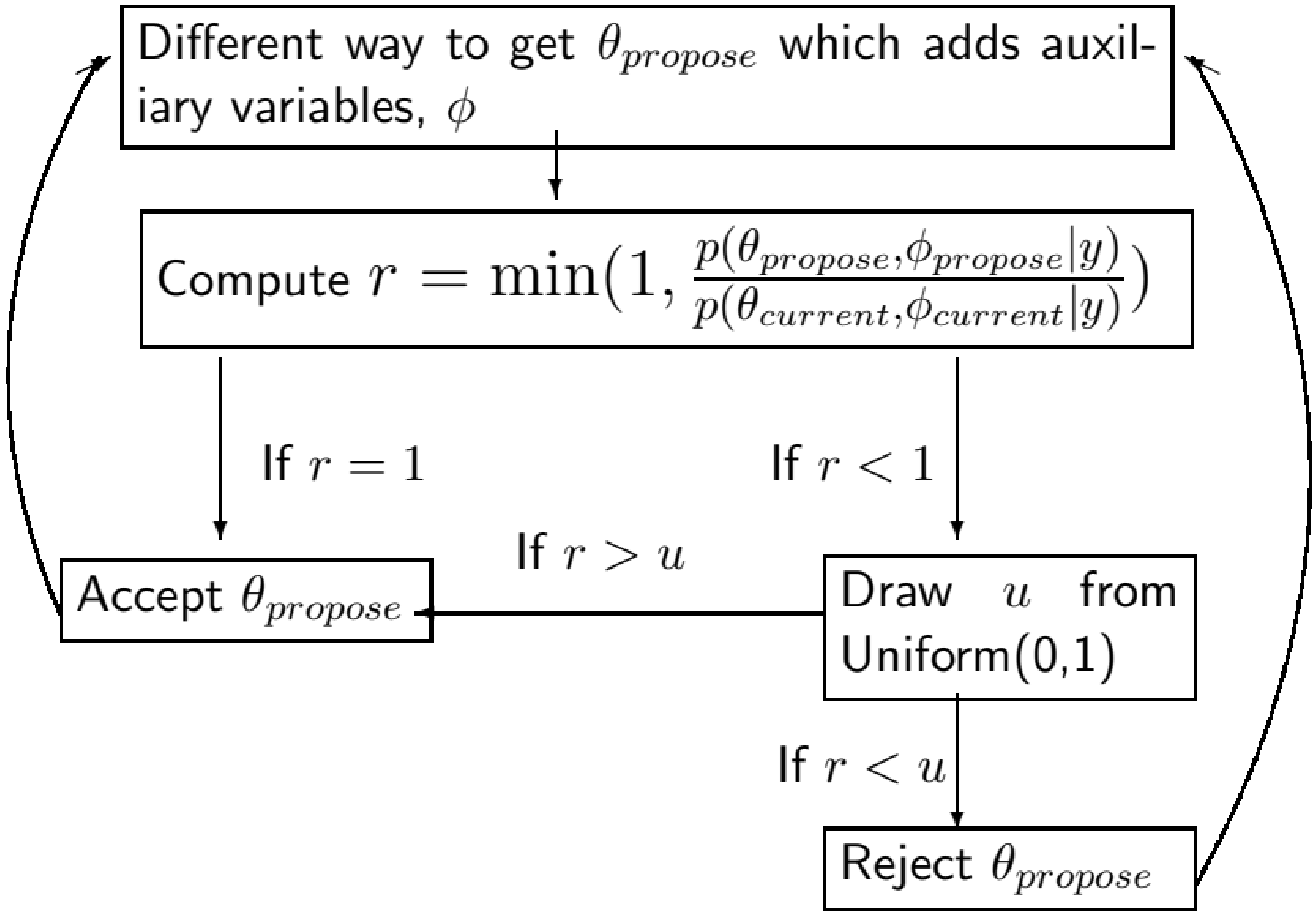
3 steps of the HMC algorithm

Start with an initial parameter value θ and assign random values to the momentum variable m, usually drawn from a standard normal distribution

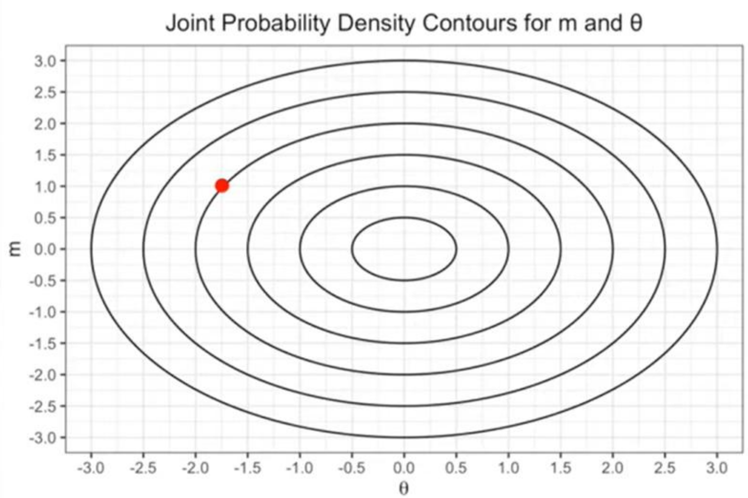
Using the Hamiltonian equations of motion, simulate the trajectory of the parameter and momentum through the state space

After simulating the dynamics, accept or reject the new state based on a Metropolis acceptance criterion

$$\alpha = \min\left(1, \exp\left(H(\theta_0, m_0) - H(\theta_T, m_T)\right)\right)$$


Different way to get $\theta_{propose}$ which adds auxiliary variables, ϕ
Compute $r = \min(1, \frac{p(\theta_{propose}, \phi_{propose}|y)}{p(\theta_{current}, \phi_{current}|y)})$
If $r = 1$
If $r < 1$
If $r > u$
Accept $\theta_{propose}$
Draw u from Uniform(0,1)
If $r < u$
Reject $\theta_{propose}$

Start with an initial parameter value θ and assign random values to the momentum variable m



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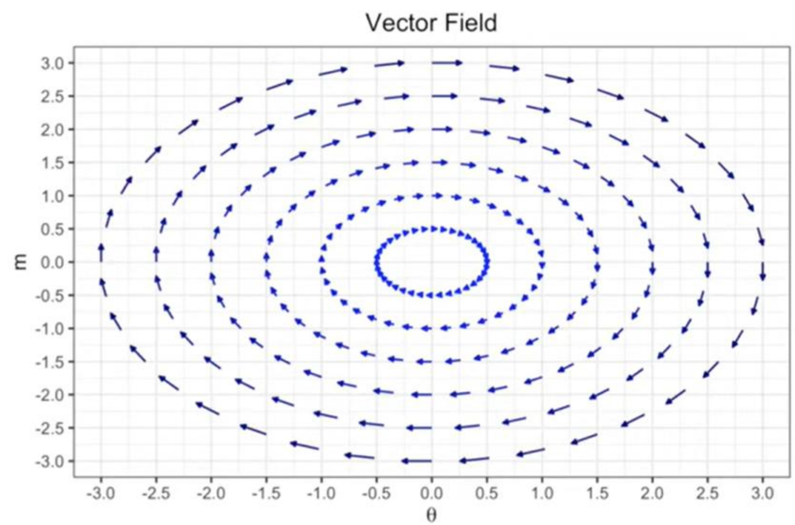
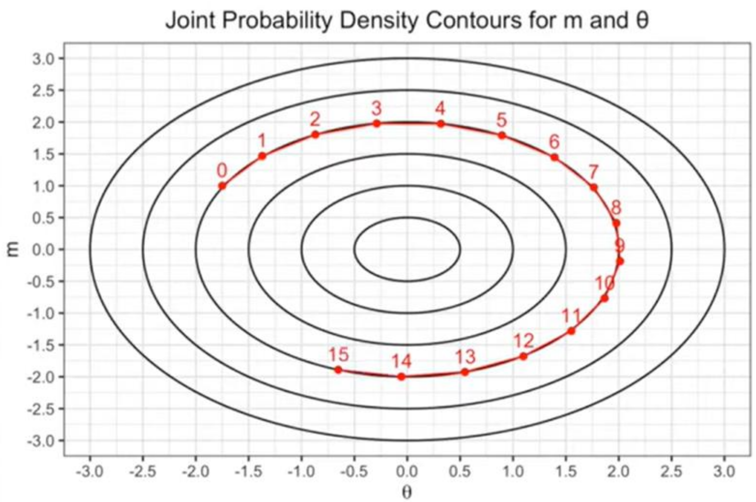


$\theta_{current}$ = -1.75, m = 1.00 (m is randomly drawn from N(0,1) distribution)

Using the Hamiltonian equations of motion, simulate the trajectory of the parameter and momentum through the state space



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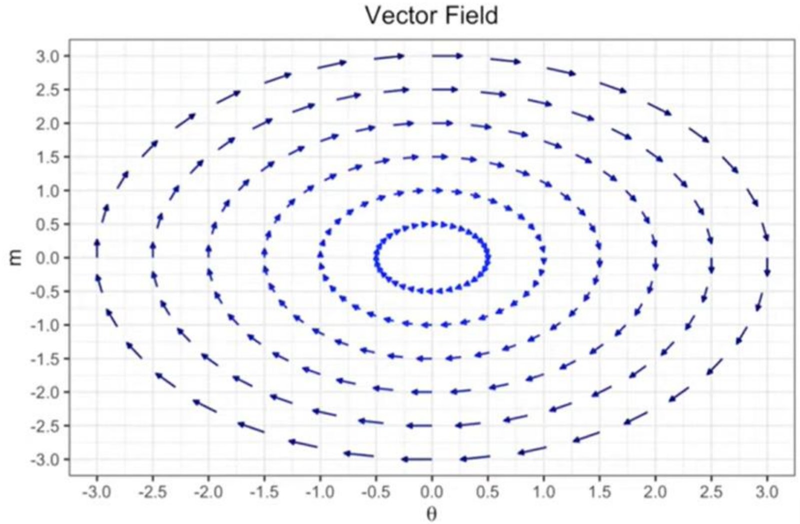
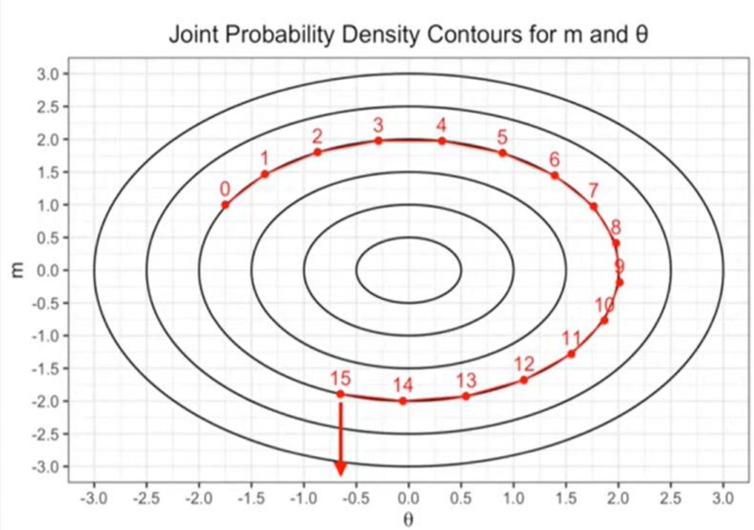


$\theta_{current} = -1.75$, $m = 1.00$

L = number of steps = 15

ε = step size = 0.3

At the end of the trajectory, only keep the new parameter value.....repeat steps 1 - 3



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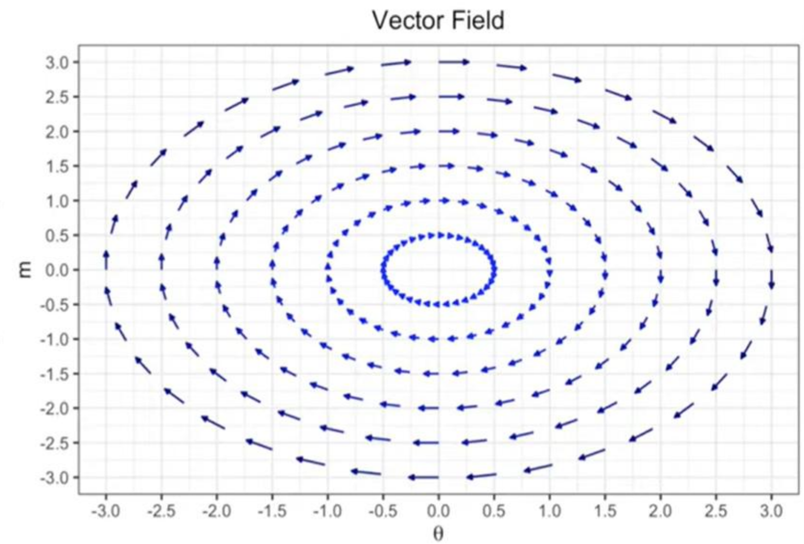


$\theta_{current}$ = -1.75, m = 1.00

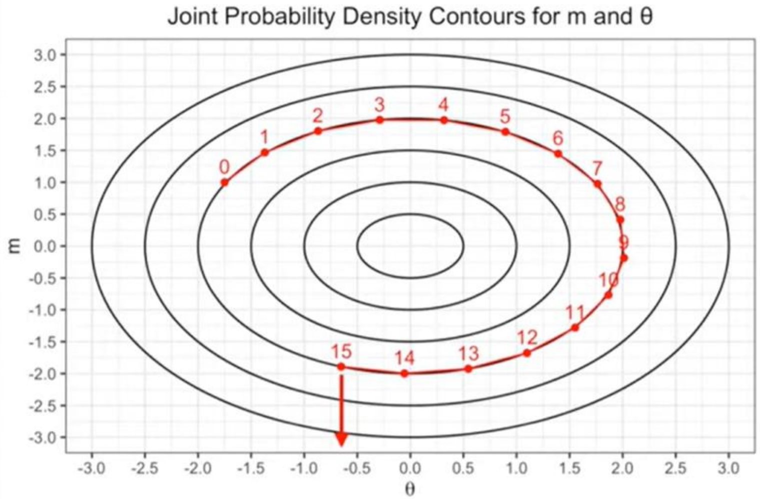
θ_{new} = -0.65

L = number of steps = 15

ε = step size = 0.3

How large should each step be (step size – ε)?

ε = 0.3



$$\frac{d\theta}{dt} = \frac{\partial H}{\partial m}$$

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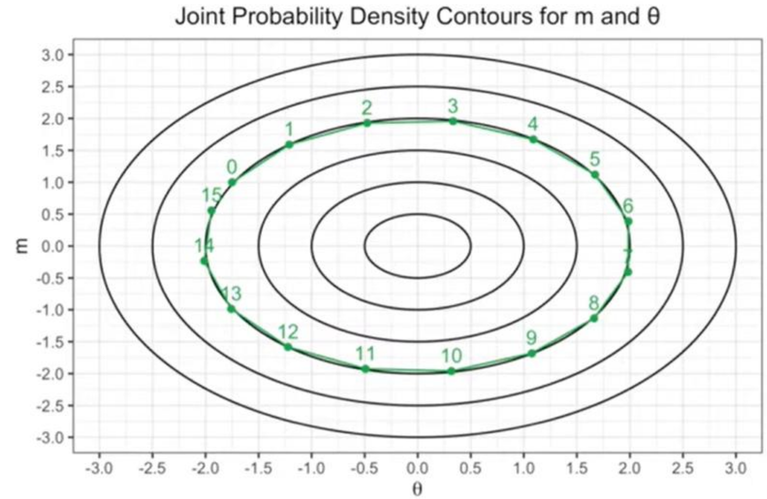
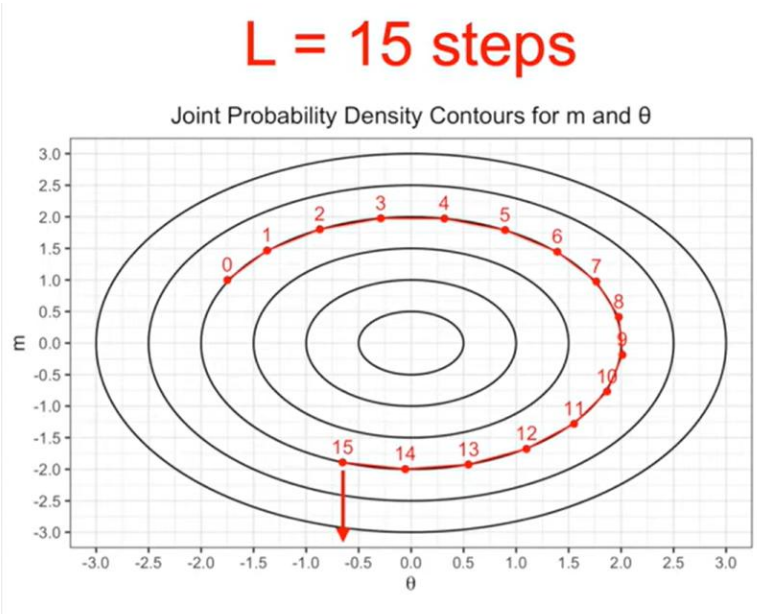
ε = 0.4



$\theta_{current}$ = -1.75, m = 1.00

L = number of steps = 15

How many steps should we do?



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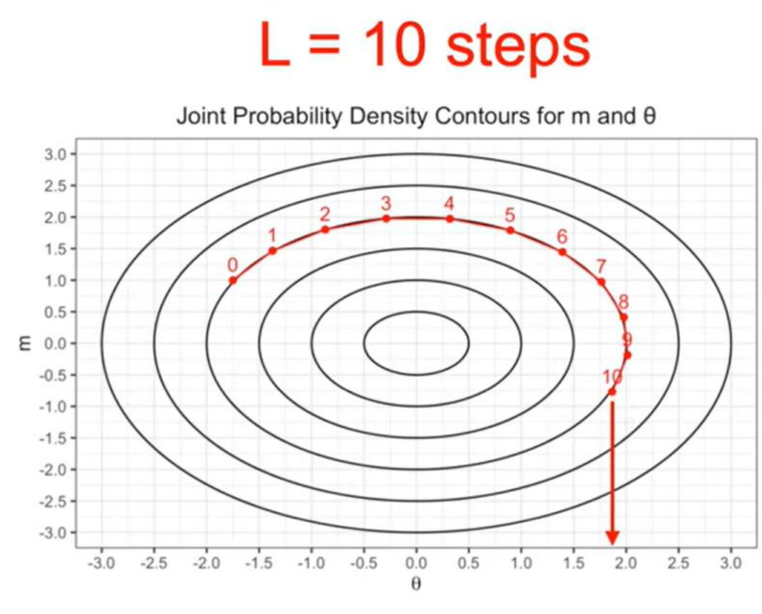


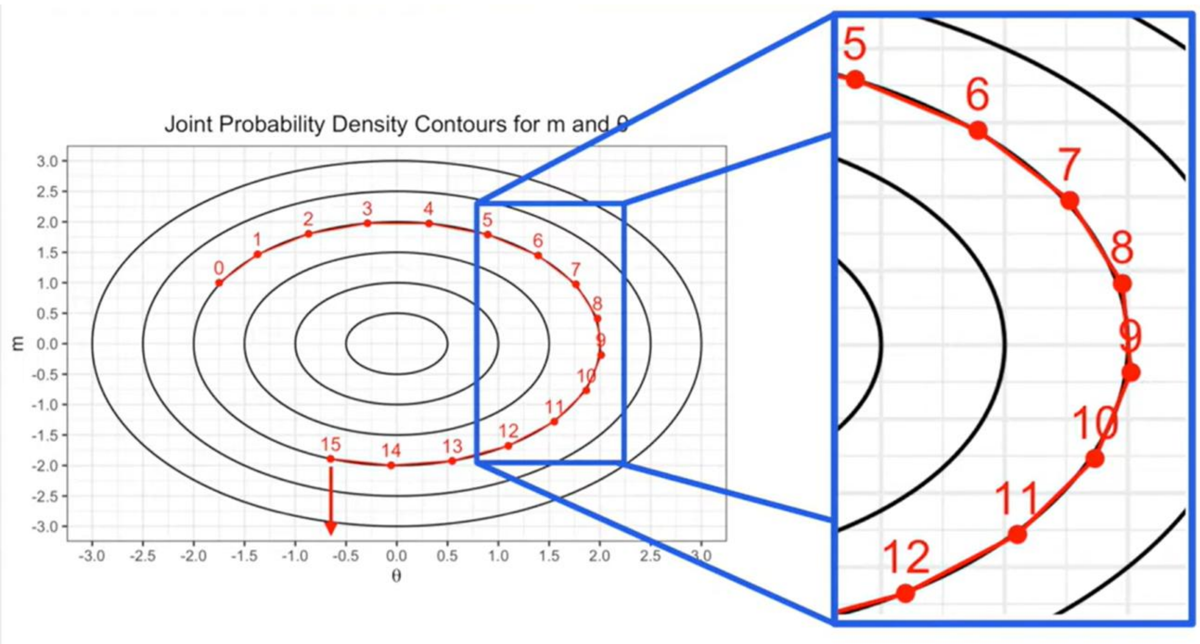
$\theta_{current} = -1.75$, $m = 1.00$

ε = step size = 0.3

Solving the differential equation (Hamiltonian equation)

- Exact analytical solution is not possible

$$\frac{d\theta}{dt} = \frac{\partial H}{\partial m}$$
$$\frac{dm}{dt} = -\frac{\partial H}{\partial \theta}$$



After step 9, we have:

$$\frac{P(\theta, m)_{proposal}}{P(\theta, m)_{current}} = 0.99$$

If we stop at point #9, we would only accept this proposal 99% of time

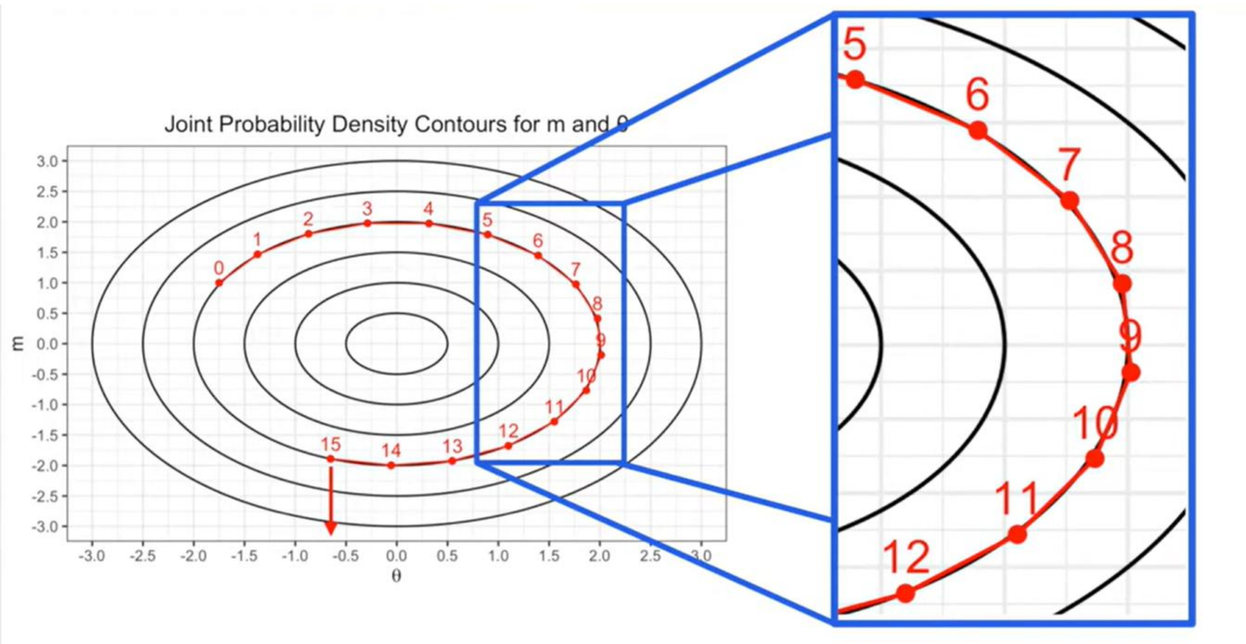
(a perfect integrator accepts proposal 100% of the time)

Solving the differential equation (Hamiltonian equation)

Leapfrog Integrator

$$\frac{d\theta}{dt} = \frac{\partial H}{\partial m}$$
$$\frac{dm}{dt} = -\frac{\partial H}{\partial \theta}$$

- Exact analytical solution is not possible



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(a perfect integrator accepts proposal 100% of the time)

$$T = \ Accept\ proposal\ with\ Probability = min\left(1, \frac{P(\theta, m)_{proposal}}{P(\theta, m)_{current}}\right)$$

More formally, accept this transition probability with minimum of the two values

This correction to transition prob. is called the ‘Metropolis update’

Solving the differential equation (Hamiltonian equation)

Leapfrog Integrator -

the leapfrog method updates both position θ and momentum m

$$m_{t+\frac{1}{2}} = m_t - \frac{\varepsilon}{2} \nabla U(\theta_t)$$

Update momentum halfway using the gradient of potential energy U(θ)

$$\theta_{t+1} = \theta_t + \frac{\varepsilon}{2} \varepsilon M^{-1} m_{t+\frac{1}{2}}$$

Update position using the full momentum

$$m_{t+1} = m_{t+\frac{1}{2}} - \frac{\varepsilon}{2} \nabla U(\theta_{t+1})$$

update momentum again to complete the full step

The Leapfrog method is symplectic, meaning it conserves important properties of the physical system, like energy and momentum, better than some other numerical methods (such as Euler's method)

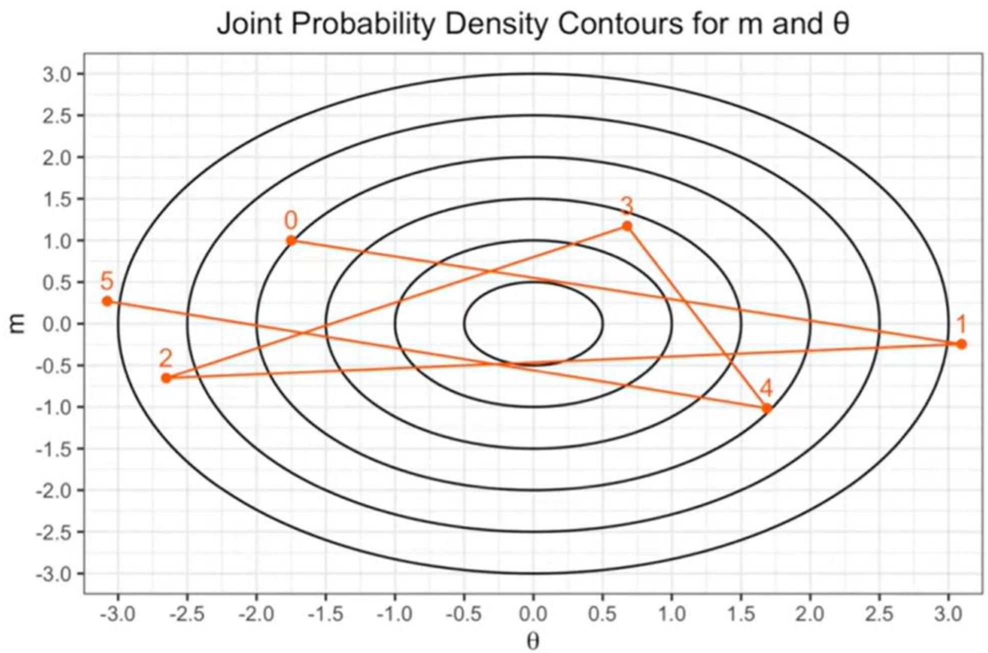
Position is the parameter of interest
In the probability space, our goal is to sample a vector of parameters θ (e.g., weights in a machine learning model). These represent the "position" in the landscape of our probability distribution

Momentum is a helper
We introduce a new variable, momentum m, to help navigate this landscape. In physics, momentum keeps a particle moving, preventing it from wandering randomly (as traditional MCMC might do)

Hamiltonian as a total energy function
HMC treats the log of the probability distribution we want to sample from, $\log p(\theta)$, as the potential energy $U(\theta)$. The momentum variable m provides the kinetic energy. Together, these define a Hamiltonian $H(\theta,m)$, which governs the system's behavior

Leapfrog method - Discrete updates
HMC uses the leapfrog integrator, a numerical method from physics, to simulate how the position and momentum evolve. This allows us to take smooth, controlled steps rather than random jumps

Leap frog method can go wrong sometime



$\theta_{current}$ = -1.75, m = 1.00 L = number of steps = 5 ε = step size = **1.85**

ε is far too large, the Hamiltonian P(θ, m), is oscillating wildly

Efficient implementation of HMC require careful optimization of step size (ε) and number of steps (L)

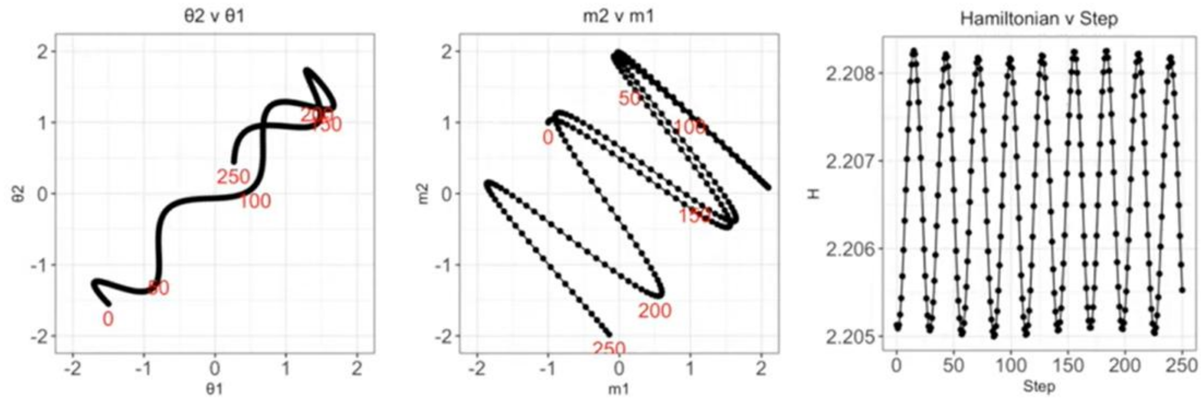
Demonstration of leapfrog – https://observablehq.com/@herbps10/leapfrog-integrator

θ = -1.50, -1.55 ρ = 0.95

m = -1.00, 1.00 (correlation)

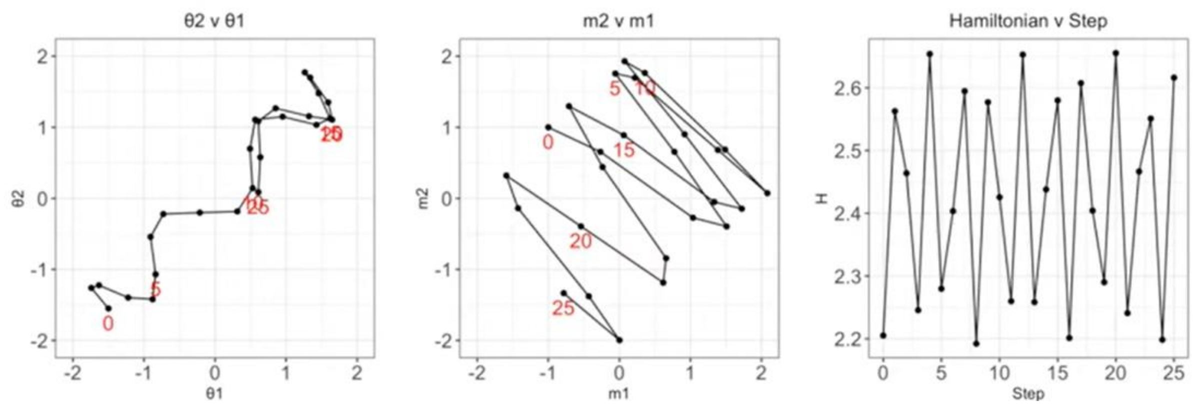
L = 250

ε = 0.025

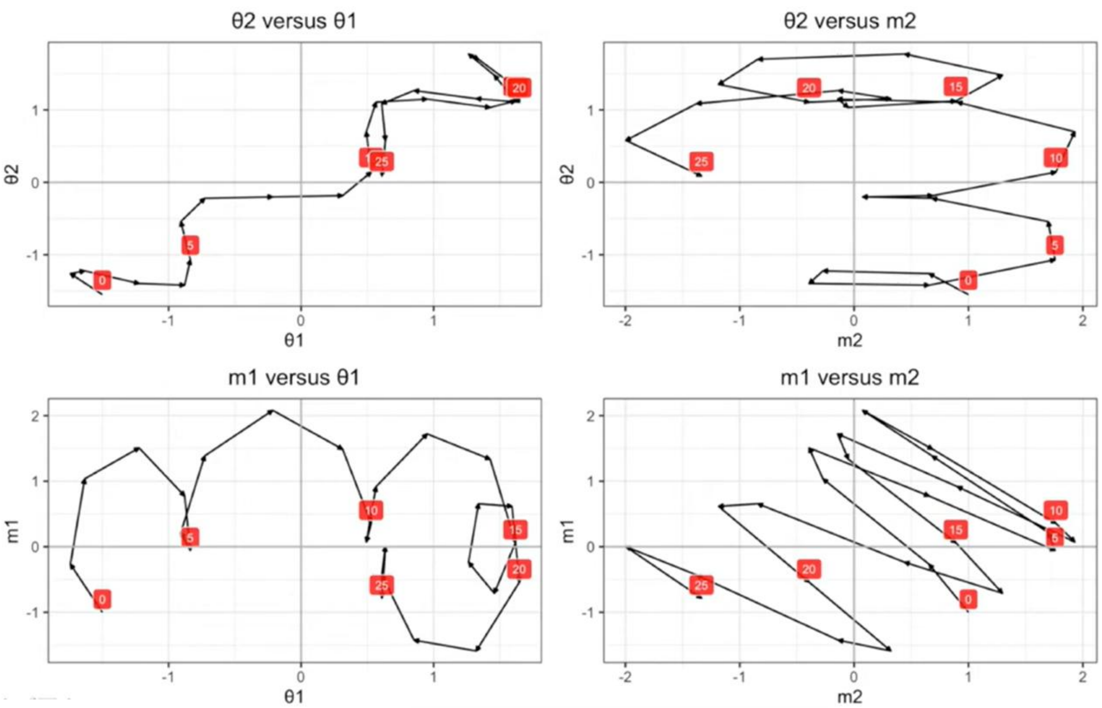


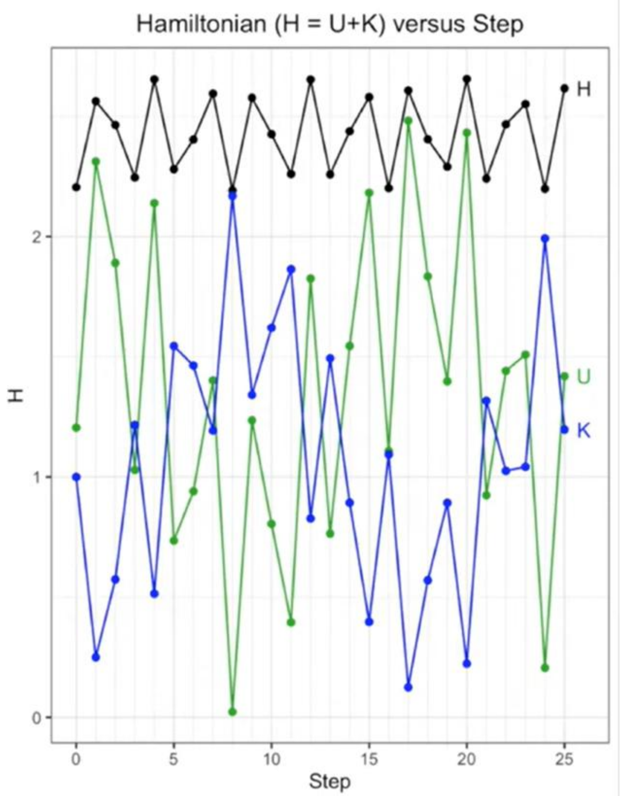
L = 25

ε = 0.25



Contribution of the potential (U) and Kinetic (K) energy to the overall value of the Hamiltonian





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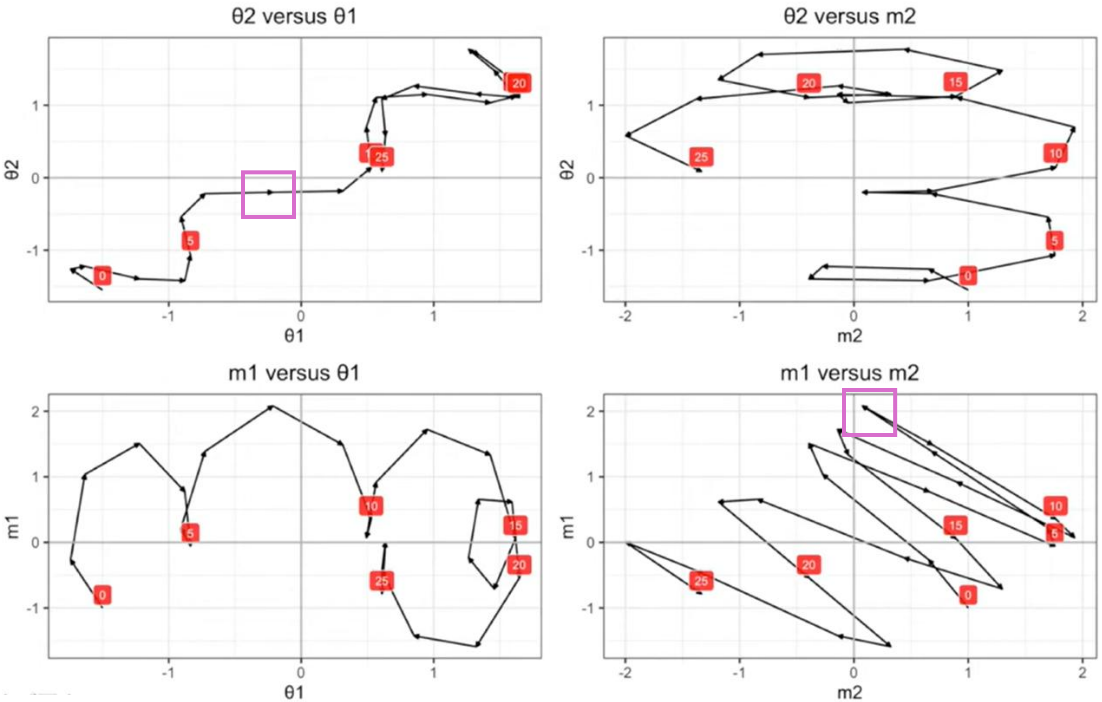

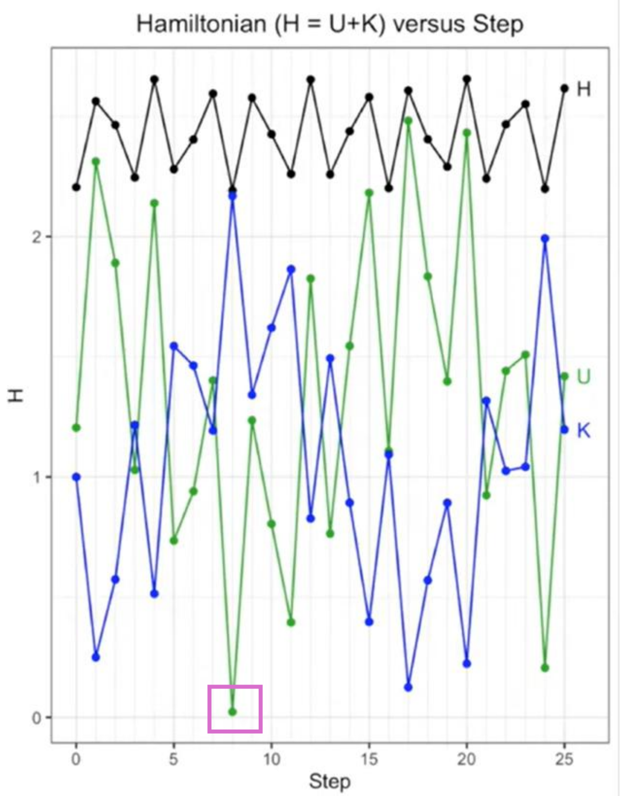

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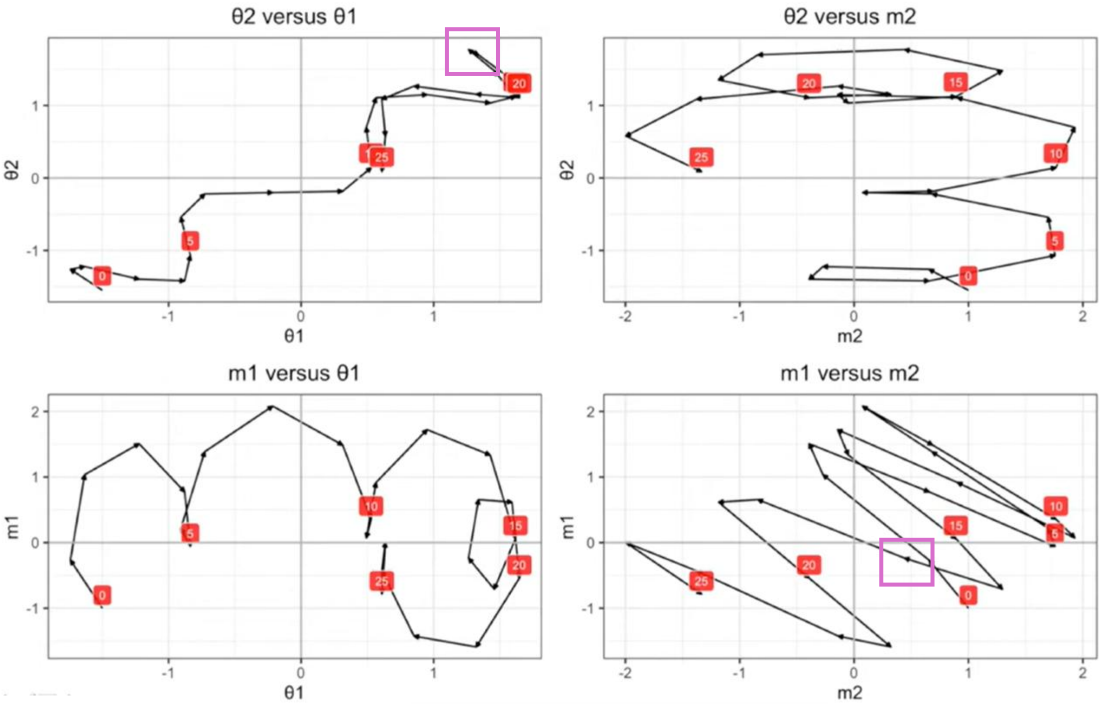
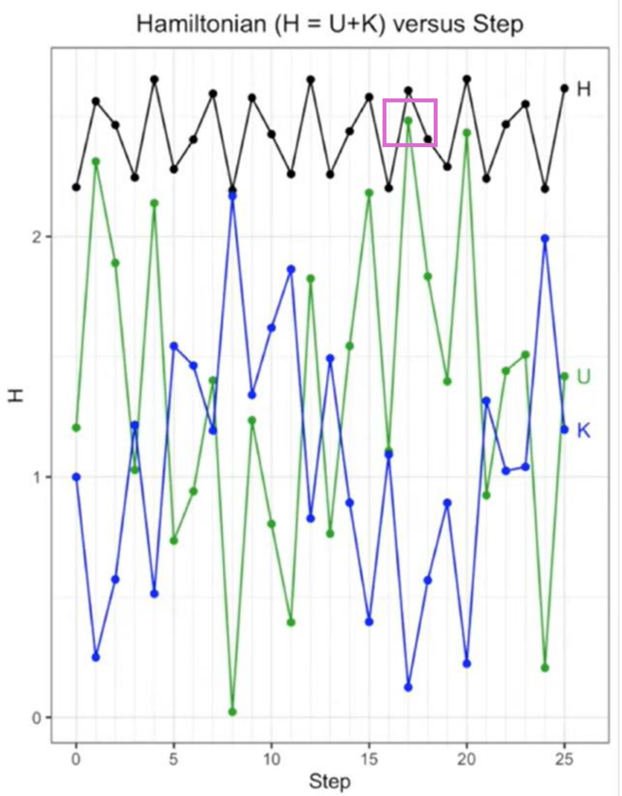
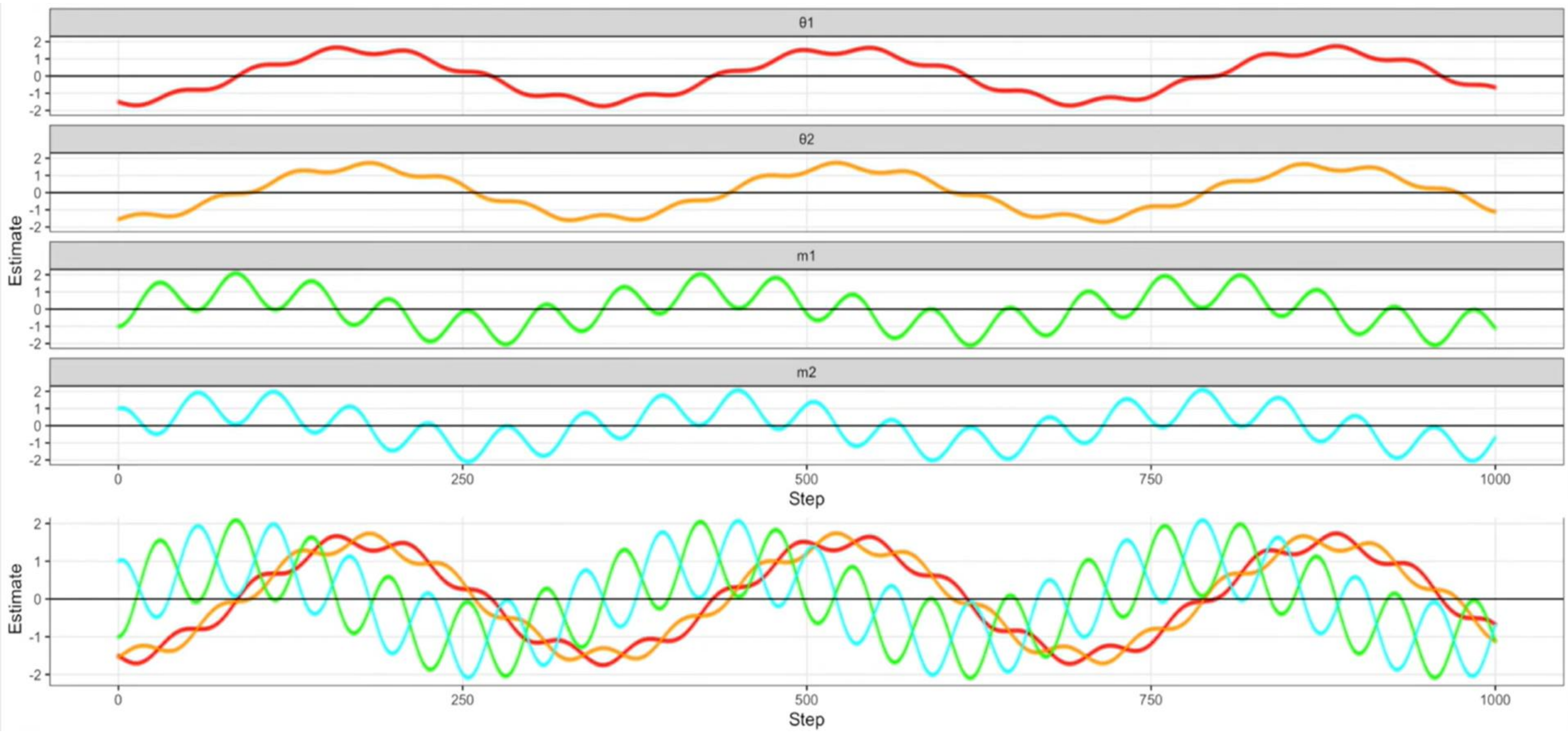




If we do 1000 steps, we see the cyclic nature of HMC



Some final notes on HMC

- Basic HMC shown here is the foundation, there are variations that can be implemented based on the problem at hand
- Trajectory lengths can be optimized (e.g. the NUTS – No U-turn Sampler – algorithm samples backwards and forwards in time until we double back on ourselves (in parameter space))
- Need to scale our parameter/momentum variables to ensure good performance
- Automatic differentiation ensures fast/accurate gradient calculations
- As a tool for Bayesian estimation of complex models, HMC looks great